The worst thesis in the world

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ABSTRACT

[This abstract must provide a succinct and informative condensation of your work. Candidates are welcome to prepare a lengthier abstract for inclusion in the dissertation, and provide a shorter one in the CaltechTHESIS record.]

PUBLISHED CONTENT AND CONTRIBUTIONS

First author contributions: [20], [14]

Major contributions: [6], [15], [21]

SpECTRE core developer and maintainer of Cauchy-Characteristic Extraction (CCE) module: [11], [12], [22], [19], [3], [8], [9], [17], [4], [10], [7], [23], [16], [18], [2], [1], [13], [24], [25], [5]

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Chapter 1

INFRASTRUCTURE FOR CROSS-CODE COMPARISON OF CCE WAVEFORMS

1. Motivation
    a) Explain importance about comparing waveforms
    b) Waveform models, surrogates, etc.
    c) h_+ and $h_\times$ not gauge invariant
    d) Analog to 4 potential from E&M
    e) Different evolution gauges
2. Overview of CCE (from Jordan's paper)
3. Different formulations evolved variable
4. Converting to "standard" variables for CCE
5. Caveat of frame fixing/BMS transforms

1.1 Introduction

Being able to compare gravitational waveforms with each other, whether it be comparing Numerical Relativity (NR) waveforms with other NR waveforms or observed gravitational waves (GWs), is an extremely important ability in GW science; potentially the most important. GW detectors compare suspected signals to models that have been informed by NR such as the IMRPhenom family (examples: [1–3]) for frequency-domain analysis and the SEOBNR family (examples: [4–6]) for time-domain analysis. NR codes also compare waveforms internally, typically for verification of convergence as in §**??** or for analysis of trends in GW catalogs [7]. However, comparing waveforms from different NR simulations, even from the same code, have to be done carefully since different simulations will be done in different gauges.

1.1.1 Electricity and Magnetism example

To give an example of why we need to consider gauges when comparing NR waveforms, let us first look at a more familiar use of gauges in an introductory

setting such as electricity and magnetism. Maxwell's equations (in vector form) are

$$\nabla \cdot \mathbf{E} = 4\pi\rho \qquad \nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t} \tag{1.1}$$

$$\nabla \cdot \mathbf{B} = 0 \qquad \nabla \times \mathbf{B} = \frac{1}{c}\left(4\pi\mathbf{J} + \frac{\partial \mathbf{E}}{\partial t}\right) \tag{1.2}$$

From Gauss's law for magnetism and Faraday's equation along with vector calculus identities, we can define

$$\mathbf{B} = \nabla \times \mathbf{A} \tag{1.3}$$

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t} \tag{1.4}$$

where ϕ is the electric potential and $\mathbf{A}$ is the vector potential. We notice that if we modify ϕ and $\mathbf{A}$ like so

$$\mathbf{A} \to \mathbf{A} + \nabla\psi \tag{1.5}$$

$$\phi \to \phi - \frac{\partial \psi}{\partial t} \tag{1.6}$$

for some scalar function ϕ, $\mathbf{B}$ and $\mathbf{E}$ in 1.3 and 1.4, respectively, remain unchanged. This ability to choose ϕ and $\mathbf{A}$ is called gauge freedom, and the choice that we make is considered the gauge.

In electricity and magnetism there are two common gauges; Coulomb and Lorenz. The Coulomb gauge is defined by

$$\nabla \cdot \mathbf{A} = 0 \tag{1.7}$$

To describe the Lorenz gauge, we combine ϕ and $\mathbf{A}$ as a four-vector

$$A^\mu = (\phi/c, \mathbf{A})\,. \tag{1.8}$$

The Lorenz gauge is then

$$\partial_\mu A^\mu = 0. \tag{1.9}$$

This differs from the Coulomb gauge because of the $\partial_0 A^0 \sim \partial_t \phi$ part. The implications that this choice of gauge has on the equations can be found in virtually any electricity and magnetism textbook [TODO:CITE]. The important point for this discussion is that even though $\mathbf{E}$ and $\mathbf{B}$ mathematically remain unchanged depending on which gauge is chosen, the gauge dependent quantities ϕ and $\mathbf{A}$ cannot be compared between gauges.

1.1.2 General and Numerical Relativity gauges

Now we move to gauges in general relativity (GR). Instead of putting a constraint on the four-potential as our gauge condition, in GR we impose our gauge condition on the coordinates x^μ of our spacetime. We skip a derivation of *why* we have this coordinate freedom in the theory of general relativity as that's the topic of introductory GR courses, but a derivation can be found in virtually any GR text book, including [8–10] to name a few. For our purposes, let's take a look at a few gauges in GR and NR.

1.1.2.1 Transverse-traceless gauge

The first gauge most students encounter is one called the transverse-traceless (TT) gauge, occasionally called the radiation gauge. The TT gauge is particularly useful in a linearized theory of gravity where we define our metric as

$$g_{\mu\nu} = \eta_{\mu\nu} + \bar{h}_{\mu\nu} \tag{1.10}$$

with

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h \tag{1.11}$$

where $\eta_{\mu\nu}$ is our flat background spacetime metric and $h_{\mu\nu}$ is a small perturbation of this background. Here, $\bar{h}_{\mu\nu}$ is a "trace-reversed" form. We take our field equations

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi T_{\mu\nu} \tag{1.12}$$

and substitute Eq. 1.10 into them. This will contain second derivatives of $\bar{h}_{\mu\nu}$ because of the Ricci tensor and scalar. It is at this point where we make a gauge choice of

$$\Box x^\mu = 0 \tag{1.13}$$

which can be shown to simplify to [10]

$$\Box x^\mu = 0 \rightarrow g^{\mu\nu}\Gamma^\rho_{\mu\nu} = 0 \rightarrow \partial_\mu \bar{h}^\mu_\lambda = 0 \tag{1.14}$$

in the weak-field (linear) limit. Plugging Eq. 1.14 into Eq. 1.12 and taking the vacuum case ($R_{\mu\nu} = T_{\mu\nu} = 0$), we arrive at a clean result

$$\Box \bar{h}_{\mu\nu} = 0. \tag{1.15}$$

It can be shown (through some math that is unimportant to the point here) [10] that

$$h^{\text{TT}}_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{1.16}$$

where we have dropped the bars because $h_{\mu\nu}$ is the trace-reverse of $\bar{h}_{\mu\nu}$ which is traceless. These are the familiar GW polarizations h_+ and $h_\times$ for a linear GW propagating in the z-direction. Therefore, this is a useful gauge to analyze GWs from one of the detectors (though it's quite a bit more complicated than this in reality [11–14]).

1.1.2.2 NR gauges

Moving on to NR, the two most popular formulations of the field equations (Eq. 1.12) are the Generalized Harmonic (GH) formulation [15] and the BSSN/Z4 family [TODO:CITE]. Each of these formulations use different gauge conditions for the evolution of the coordinates x^μ, none that are equal to the TT gauge.

We start with GH. There are two dominant gauges used when evolving the GH formulation. The first is actually the harmonic gauge introduced in Eq. 1.13. The TT gauge is actually a sub-category of the harmonic gauge, useful for linear theories. However, GH is the fully non-linear theory of gravity and so the TT gauge is not suitable here. The harmonic gauge is convenient because the solutions for the equations of the coordinates are harmonic (or wave) solutions, hence the name, and we know how to solve for waves.

The second gauge commonly used in GH evolutions is one that actually builds off of the harmonic gauge, because it's wave solutions are so desirable. The general form of this gauge, called the damped-harmonic gauge, is

$$\Box x_\mu = H_\mu(x^\nu, g_{\nu\kappa}) \tag{1.17}$$

where H_μ is called the gauge-source function [16], and it is a function of the coordinates and the spacetime metric. The gauge source function is defined in Eq. **??–??**. The main purpose for this is to damp away oscillations shift α and drive it to one (so the time evolution remains steady).

For the BSSN/Z4 family of formulations, a gauge condition called the moving-puncture gauge, combined with $1 + \log$ slicing and a Γ-driver condition, is used. For this family, conditions are put onto the lapse function α and the shift vector β^i instead of having equations for the coordinates themselves. The $1 + \log$ slicing condition is (Ch. 4 of [17])

$$\partial_t \alpha = -2\alpha K + \beta^j \partial_j \alpha \tag{1.18}$$

where K is the extrinsic curvature. The Γ-driver condition can come in two forms (Ch. 4 of [17])

$$\partial_t \beta^i = \eta B^i + \beta^j \partial_j \beta^i, \tag{1.19}$$

or ([18])

$$\partial_t \beta^i = \tilde{\Gamma}^i - \eta \beta^i + \beta^j \partial_j \beta^i, \tag{1.20}$$

where η is a tunable parameter, and $\tilde{\Gamma}^i$ is related to the Christoffel symbols. The former is typically used with BSSN while the later is used with the Z4 family, but many NR codes make slight modifications to any of these gauge equations.

1.2 CCE?

Now that we have detailed a number of different useful gauges, namely the TT, harmonic, damped harmonic, and moving-puncture gauges, it should be clear that the GW strains h we compute from simulations done in these different gauges cannot directly be compared with each other. The coordinate freedom of GR implies that even if we use the same formulas to compute h, we will get different answers because the coordinates are completely different. Therefore, how can we compare waveforms across codes?

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